CS177 Bioinformatics: Software Development and Applications
SIST, ShanghaiTech University

Spring, 2025

Homework 2

Due time: 11:59 pm, April 23, 2025 (Beijing Time).
Submission: In Blackboard, upload your solution in a zip file, including a PDF file with the written solution and Jupyter notebooks with all the outputs, etc.
Note: Please name your zip file as "CS177-HW2-your name-your student ID", for example, "CS177-HW2-JieZheng-2024233187.zip".

The questions below are assigned 40 points in total, which will constitute 10% of the final grade.

Q1. Graph Convolutional Network (GCN) on a protein-protein interaction (PPI) network (15 points)

Objective: Predicting new **PPIs** as link prediction with a GCN.

Problem Description: In this problem, you need to process the PPI data properly and then use a GCN to do link predictions. You need to complete the Python code in the attached Jupyter notebook (`HW2_ppi.ipynb`) to process the PPI data. You are expected to complete the following major components in the notebook to build the end-to-end pipeline for training and evaluating a GCN model for link prediction.

You should finish the following parts (with indicative points):

1. Implementation of data splitting. (2 pts)
2. Transforming the given data into the graph from a PPI network. (2 pts)
3. Construction and normalization of the adjacency matrix. (3 pts)
4. Implementation of the graph convolutional layer. (2 pts)
5. Construction of the `GcnNet` model. (2 pts)
6. Training and evaluation of the model. (4 pts)

Your final model performance should be reasonable. Points will be awarded based on code correctness, structure, and results.

Q2. Reproduce the BLOSUM Matrix from BLOCKS (25 points)

Objective: Reconstruct a BLOSUM substitution matrix using sequence alignment blocks from the BLOCKS database.

Problem Description: You are provided with a subset of the BLOCKS database containing multiple sequence alignments. The goal is to reproduce the BLOSUM matrix from these alignments by counting pairwise residue frequencies, clustering sequences based on similarity thresholds, and computing log-odds substitution scores. Implement the logic in the attached Jupyter notebook (`HW2_blosum.ipynb`).

You can refer to the original BLOSUM paper for background and methodology: *Amino acid substitution matrices from protein blocks* (Henikoff & Henikoff, 1992).
Available online: `https://pmc.ncbi.nlm.nih.gov/articles/PMC50453/`

You should finish the following parts:

1. Implement a function to calculate sequence identity.
2. Cluster sequences: group sequences with identity above the given threshold (e.g., 62%).
3. Count aligned residue pairs between clusters.
4. Compute log-odds scores and normalize then to build the substitution matrix.

Evaluation Criteria:

- If your final matrix built by running your code has Manhattan distance to the reference matrix ≤ 90, you receive full credit.
- If the distance is in the range $[91, 140]$, you receive 70% of the total score.
- If the distance is in the range $[141, 180]$, you receive 40% of the total score.
- If the distance is greater than 180, you receive no points.

Note: We will manually check your code to ensure that the matrix is constructed in a correct and principled way.

END OF CS177 HOMEWORK 2